Assignment 6 - memory management

7.11.2023

100%

Poengsum frakoblet modus:
100%

Legg til kommentar

Anonym vurdering: **nei**

15.12.2023

Detaljer

Introduction

Note: This assignment covers both the *pointers* and *memory management* lecture. Most students should not attempt this assignment until after the *memory management* lecture. Practice understanding and using pointers until then.

6.1

Folder name: ikt101g23h/assignments/solutions/assignment_6_1

Specification:
Write an application that:

1. Asks the user how many integers the user will type in
2. Allocates memory for the number of integers that were specified by the user
3. Reads in all the integers
4. Prints the integers in the order they were entered
5. Prints the integers in ascending order (sorted)

Requirements:
malloc() and free() must be used. You are not allowed to use a pre-allocated static size array for the values.

Expected output with no numbers:
No numbers were given

Expected output otherwise (with the right values):
Count: 5
Numbers: 4 2 1 3 5
Sorted: 1 2 3 4 5

6.2

Folder name: ikt101g23h/assignments/solutions/assignment_6_2

Specification:
Write an application that asks for student information (name and age) until the name given is stop, then:

- Prints out how many students that are stored
- Prints out all the collected student information
- Prints out the name of the youngest student
- prints out the name of the oldest student

Requirements:
The student info must be stored in a struct.
malloc(), realloc() and free() must be used. You are not allowed to use a pre-allocated static size array for the student list. Grow your array when more space is needed.

Expected output with no students:
No students were given

Expected output otherwise (with the values that were typed in, of course):
Count: 3
Name = Arne Banan, Age = 23
Name = Lisa Potet, Age = 22
Name = Knut Appelsin, Age = 24
Youngest: Lisa Potet
Oldest: Knut Appelsin

Test locally

see *Local testing with Docker* under Assignments.

Test the solution on your own computer. You can fix your code and retest as many times as you want.

Deliver and test in Bamboo

See *Delivery and Bamboo testing* under Assignments.

Finally deliver and test the solution in Bamboo. This is what updates your score in Canvas which is the score I can see. You can test as many times as you want here as well up until the final deadline.